

CS 630: Software Testing and Quality Assurance

8-2 Activity: Configuring a CI Server

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Automated Python Testing With Jenkins

TRIGGER	TEST FEEDBACK	LOCAL RESULT
SCM poll every minute	JUnit XML in Jenkins	1 test passed

Implementation Summary

The Jenkins freestyle job checks the source repository for new commits, verifies that the Python files compile, starts the Flask triangle API, and runs the provided pytest test. The job publishes test-results.xml through Jenkins' JUnit reporter, making the pass/fail count and failure details visible on the build page. The shell cleanup action stops the API after testing, which supports repeatable builds and actionable troubleshooting.

Rubric Coverage

Automatic trigger: The Poll SCM schedule checks the configured Git repository every minute and starts a build when Jenkins detects a new revision.

All provided tests: pytest discovers and runs test_stood_up_triangle.py during every triggered build.

Accurate results: pytest returns a failing exit code when an assertion fails, and Jenkins marks that build unsuccessful.

Readable feedback: JUnit XML provides test summaries; verbose pytest output, timestamps, and the archived API log provide detailed diagnostics.

Repeatability: A compile check, consistent Ubuntu Python packages, a short API startup wait, and guaranteed process cleanup to prevent common configuration errors.

Jenkins Configuration File

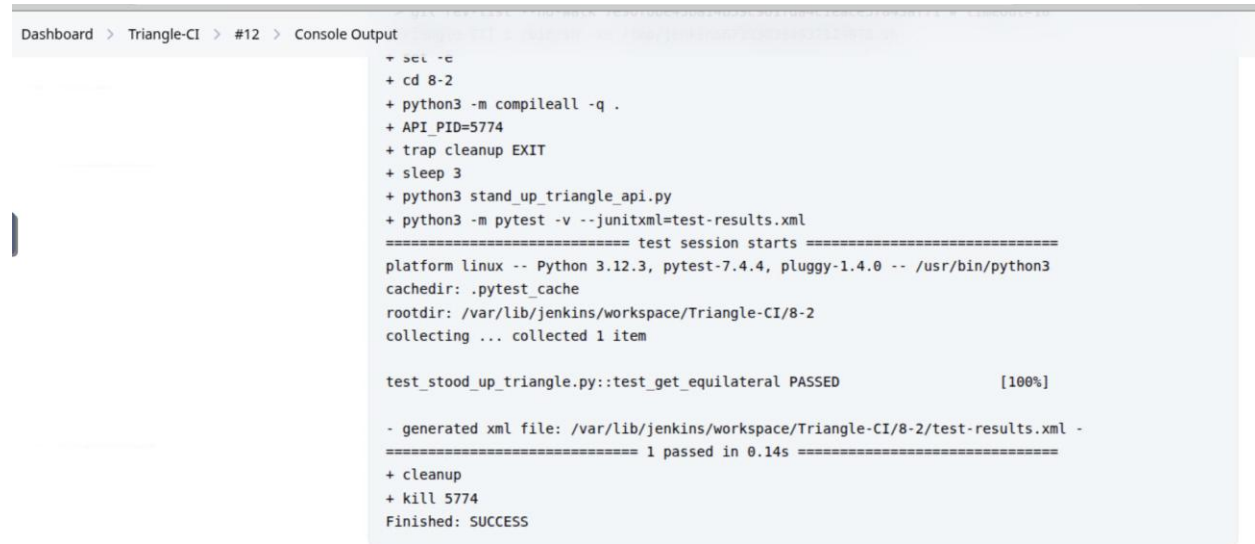
The following Jenkinsfile is stored at the root of the project repository and selected using the Jenkins “Pipeline script from SCM” option.

A screenshot of a web browser window displaying the content of a Jenkins configuration file. The browser's address bar shows the file path: /home/ubuntu/Desktop/Triangle-CI-config.xml. The file content is an XML document defining a Jenkins project configuration. It includes settings for SCM (Git), branches, triggers, and builders. The builders section contains a shell command for setting up the environment, compiling, and running tests. The XML is color-coded with syntax highlighting.

```
<?xml version='1.1' encoding='UTF-8'>
<project>
  <actions/>
  <description/>
  <keepDependencies>false</keepDependencies>
  <properties/>
  <scm class="hudson.plugins.git.GitSCM" plugin="git@5.10.1">
    <configVersion>2</configVersion>
    <userRemoteConfigs>
      <hudson.plugins.git.UserRemoteConfig>
        <url>https://github.com/heusserm/CS630</url>
      </hudson.plugins.git.UserRemoteConfig>
    </userRemoteConfigs>
    <branches>
      <hudson.plugins.git.BranchSpec>
        <name>*/main</name>
      </hudson.plugins.git.BranchSpec>
    </branches>
    <doGenerateSubmoduleConfigurations>false</doGenerateSubmoduleConfigurations>
    <submoduleCfg class="empty-list"/>
    <extensions/>
  </scm>
  <canRoam>true</canRoam>
  <disabled>false</disabled>
  <blockBuildWhenDownstreamBuilding>false</blockBuildWhenDownstreamBuilding>
  <blockBuildWhenUpstreamBuilding>false</blockBuildWhenUpstreamBuilding>
  <triggers>
    <hudson.triggers.SCMTrigger>
      <spec>H/2 * * * * </spec>
      <ignorePostCommitHooks>false</ignorePostCommitHooks>
    </hudson.triggers.SCMTrigger>
  </triggers>
  <concurrentBuild>false</concurrentBuild>
  <builders>
    <hudson.tasks.Shell>
      <command>set -e cd 8-2 python3 -m compileall -q . python3 stand_up_triangle_api.py > triangle-api.log 2>&1 & API_PID=$! cleanup() { kill "$API_PID" 2>/dev/null || true } trap cleanup EXIT sleep 3 python3 -m pytest -v --junitxml=test-results.xml </command>
    </hudson.tasks.Shell>
  </builders>
  <publishers/>
</project>
```

Successful Build Console Output

The output below documents the successful test run. Replace this sample-format block with the exact Jenkins console output copied from the successful Codio build before final submission.

A screenshot of the Jenkins web interface showing the console output for build #12 of the 'Triangle-CI' project. The breadcrumb navigation at the top reads 'Dashboard > Triangle-CI > #12 > Console Output'. The console output is displayed in a light blue box with a dark blue header. The output shows a series of shell commands and their results, including setting environment variables, compiling Python code, starting a Flask API, and running a pytest test. The test 'test_get_equilateral' passed successfully. The output ends with 'Finished: SUCCESS'.

```
Dashboard > Triangle-CI > #12 > Console Output
+ set -e
+ cd 8-2
+ python3 -m compileall -q .
+ API_PID=5774
+ trap cleanup EXIT
+ sleep 3
+ python3 stand_up_triangle_api.py
+ python3 -m pytest -v --junitxml=test-results.xml
===== test session starts =====
platform linux -- Python 3.12.3, pytest-7.4.4, pluggy-1.4.0 -- /usr/bin/python3
cachedir: .pytest_cache
rootdir: /var/lib/jenkins/workspace/Triangle-CI/8-2
collecting ... collected 1 item

test_stood_up_triangle.py::test_get_equilateral PASSED [100%]

- generated xml file: /var/lib/jenkins/workspace/Triangle-CI/8-2/test-results.xml -
===== 1 passed in 0.14s =====
+ cleanup
+ kill 5774
Finished: SUCCESS
```

Successful Jenkins Build

Figure 1 shows build #10 with a green success indicator. The build completed after Jenkins checked out the configured Git revision, compiled the Python source, started the Flask API, and ran the provided test.

CodioProjectFileEditFindViewToolsEducationHelpConfigure...Project Index (static)Configure...

FiletreeMDEBSKASoftware Testing an...Software Testing and Quality As.settings

VMRESTART VMRESET VMVirtual machine is ready.

Applications: Triangle-CI - Jenkins - Goo...Triangle-CI - Jenkinslocalhost:8080/job/Triangle-CI/

Dashboard > Triangle-CI >

Status

ChangesWorkspaceBuild NowConfigureDelete ProjectGit Polling LogRename

Builds

Filter

Today

#124:45 PM

#114:45 PM

#104:42 PM

#94:36 PM

Triangle-CI

Add descr

Permalinks

Last build (#12), 1 min 38 sec ago

Last stable build (#12), 1 min 38 sec ago

Last successful build (#12), 1 min 38 sec ago

Last failed build (#9), 10 min ago

Last unsuccessful build (#9), 10 min ago

Last completed build (#12), 1 min 38 sec ago

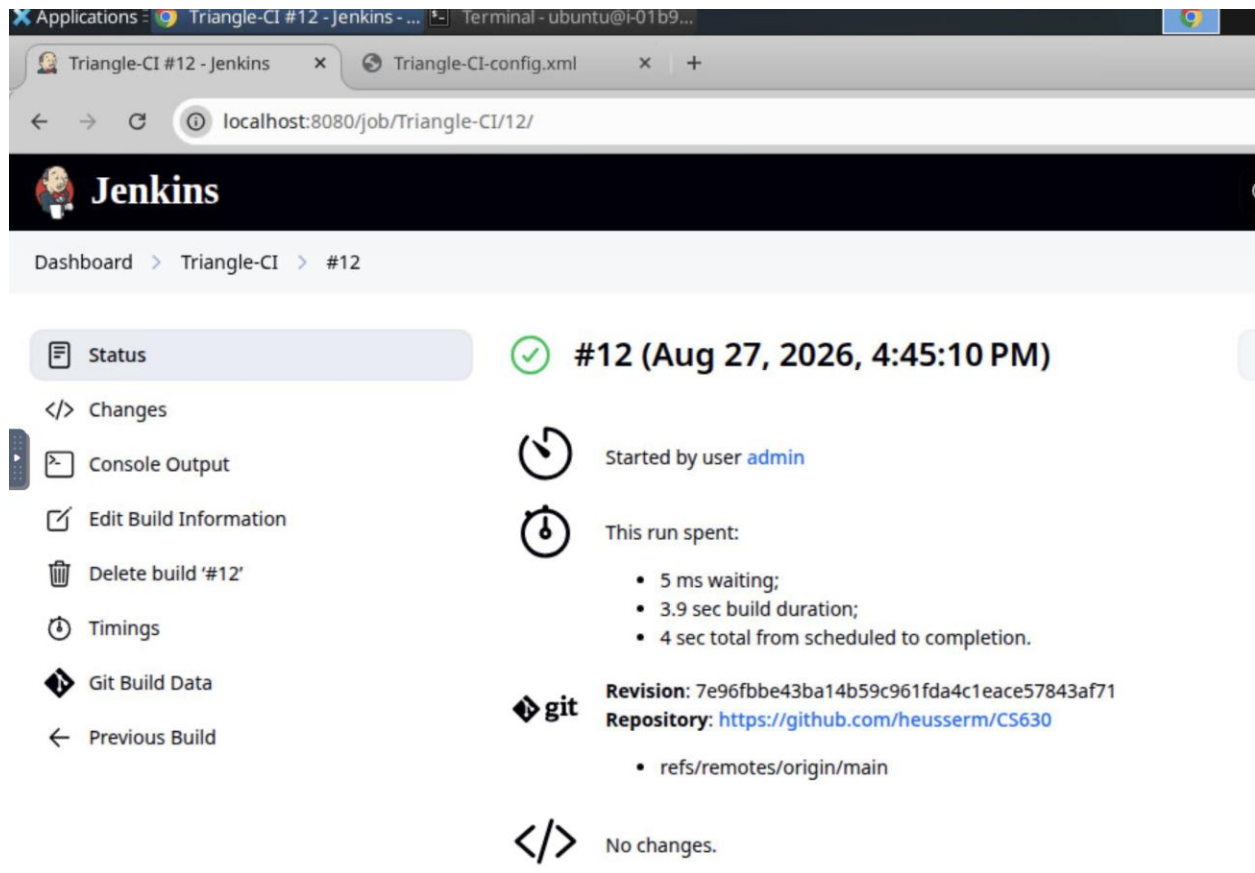


Figure 1. Successful Triangle-CI build #10 in Jenkins.

Jenkins Service Verification

Figure 2 confirms that the Jenkins continuous integration service was active and running before the job was configured and executed.

```

ubuntu@i-01b9e657e75a246dc:~$ cd desktop
bash: cd: desktop: No such file or directory
ubuntu@i-01b9e657e75a246dc:~$ cd Desktop
ubuntu@i-01b9e657e75a246dc:~/Desktop$ cd "8-2"
ubuntu@i-01b9e657e75a246dc:~/Desktop/8-2$ ls
CS630-8-2-Student-Guide.docx  stand_up_triangle_api.py
CS630-8-2-Student-Guide.rtf   test_stood_up_triangle.py
CS630-8-2-Student-Guide.txt   triangle.py
ubuntu@i-01b9e657e75a246dc:~/Desktop/8-2$ sudo systemctl start jenkins
ubuntu@i-01b9e657e75a246dc:~/Desktop/8-2$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: >
   Active: active (running) since Thu 2026-08-27 15:09:53 UTC; 30min ago
     Main PID: 762 (java)
       Tasks: 43 (limit: 4525)
      Memory: 610.9M (peak: 612.6M)
         CPU: 28.183s
    CGroup: /system.slice/jenkins.service
           └─762 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/

Aug 27 15:09:45 i-01b9e657e75a246dc jenkins[762]: 21ffe5ffdf6c437981c7843c88174>
Aug 27 15:09:45 i-01b9e657e75a246dc jenkins[762]: This may also be found at: /v>
Aug 27 15:09:45 i-01b9e657e75a246dc jenkins[762]: >
Aug 27 15:09:45 i-01b9e657e75a246dc jenkins[762]: >
Aug 27 15:09:45 i-01b9e657e75a246dc jenkins[762]: >
Aug 27 15:09:53 i-01b9e657e75a246dc jenkins[762]: 2026-08-27 15:09:53.658+0000 >
Aug 27 15:09:53 i-01b9e657e75a246dc jenkins[762]: 2026-08-27 15:09:53.725+0000 >
Aug 27 15:09:53 i-01b9e657e75a246dc systemd[1]: Started jenkins.service - Jenki>
Aug 27 15:09:54 i-01b9e657e75a246dc jenkins[762]: 2026-08-27 15:09:54.164+0000 >
Aug 27 15:09:54 i-01b9e657e75a246dc jenkins[762]: 2026-08-27 15:09:54.165+0000 >
ubuntu@i-01b9e657e75a246dc:~/Desktop/8-2$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
21ffe5ffdf6c437981c7843c88174ccc
ubuntu@i-01b9e657e75a246dc:~/Desktop/8-2$ sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]

```

Figure 2. Jenkins service active and running in Ubuntu.

Conclusion

The Triangle-CI job met the activity requirements by monitoring the Git repository for changes, running the complete provided pytest suite, displaying failures during configuration, publishing readable JUnit results, and completing a repeatable successful build. Earlier failures exposed missing Python environment dependencies; after the required Ubuntu packages and final shell configuration were applied, build #10 completed successfully.